Specifics of metadata development for qualitative social research

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In Zusammenarbeit mit

Agenda

1. Challenges for the description of qualitative data with metadata
2. Three possible approaches to improve the description of qualitative data with metadata

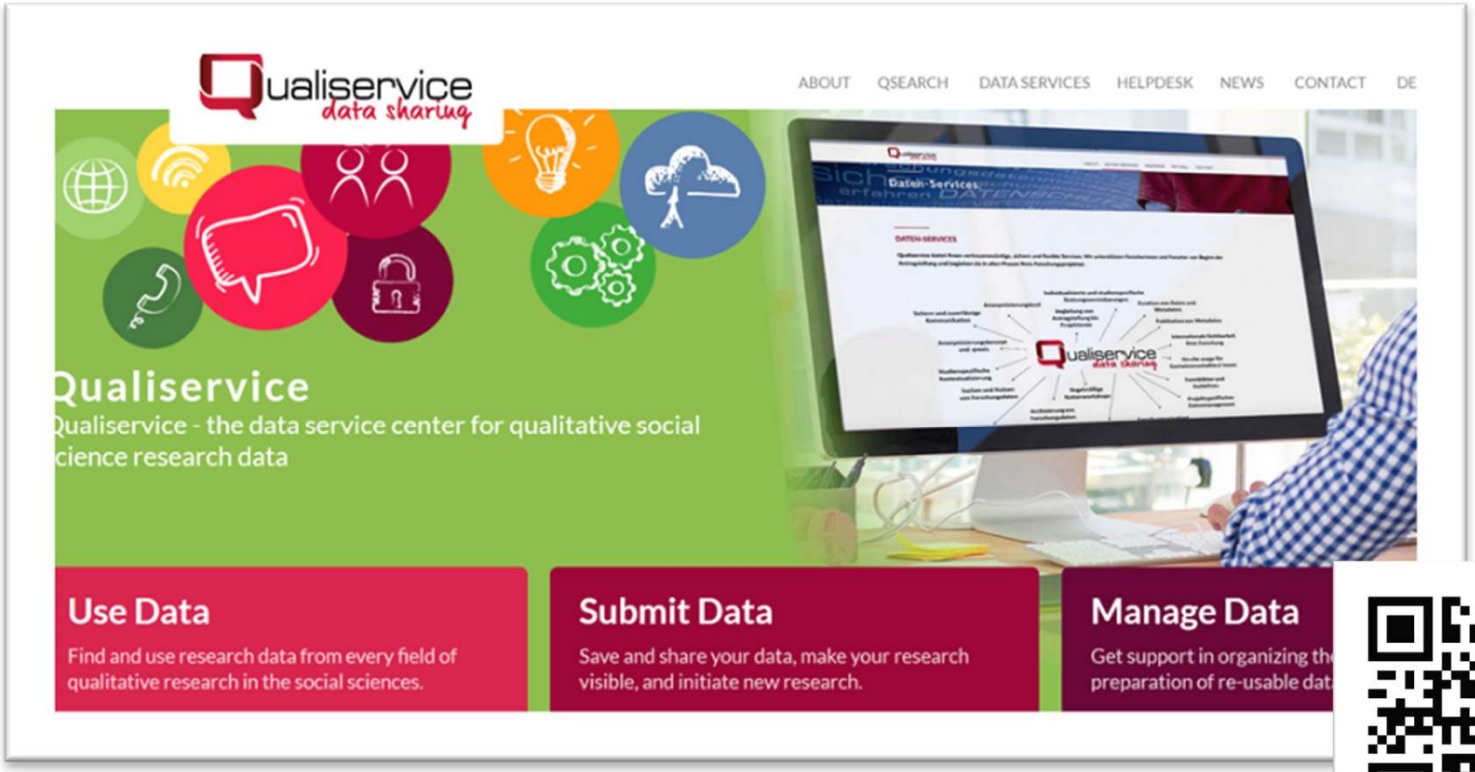
Challenges for the description with metadata

- Qualitative data is generally sensitive and often not openly accessible - metadata is particularly important
- Diversity of topics in the data material, which goes far beyond the original research project
- Qualitative research is heterogeneous, flexible metadata fields are needed to capture specific aspects that go beyond standardised fields
- Metadata schemas are developed for standardised research processes; methodological diversity and the variety of qualitative data types are not addressed in them and
- there is a lack of controlled vocabularies
- Datasets often contain a large number of cases; there is need for dataset description to retrieve information at case level

Possible approaches addressing these challenges

- The metadata-model of the RDC Qualiservice
- The controlled vocabulary QualiTerm
- International model-development in the context of the DDI-CDI Qualitative Data Subgroup

The Research Data Center Qualiservice



https://www.qualiservice.org/en/

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The Research Data Center Qualiservice

- Located at the University of Bremen, Germany
- Archiving of a wide range of qualitative data types (such as interviews, observations, field notes, audiovisual materials, photos, etc.)
- Cooperation with the world archive PANGAEA for the metadata, adaptation of the data model to the special needs of qualitative research
- Cooperation with the Bremen State and University Library for the design and further development of the metadata model, publication of contextual material to embed the dataset in its context of origin (provenance)

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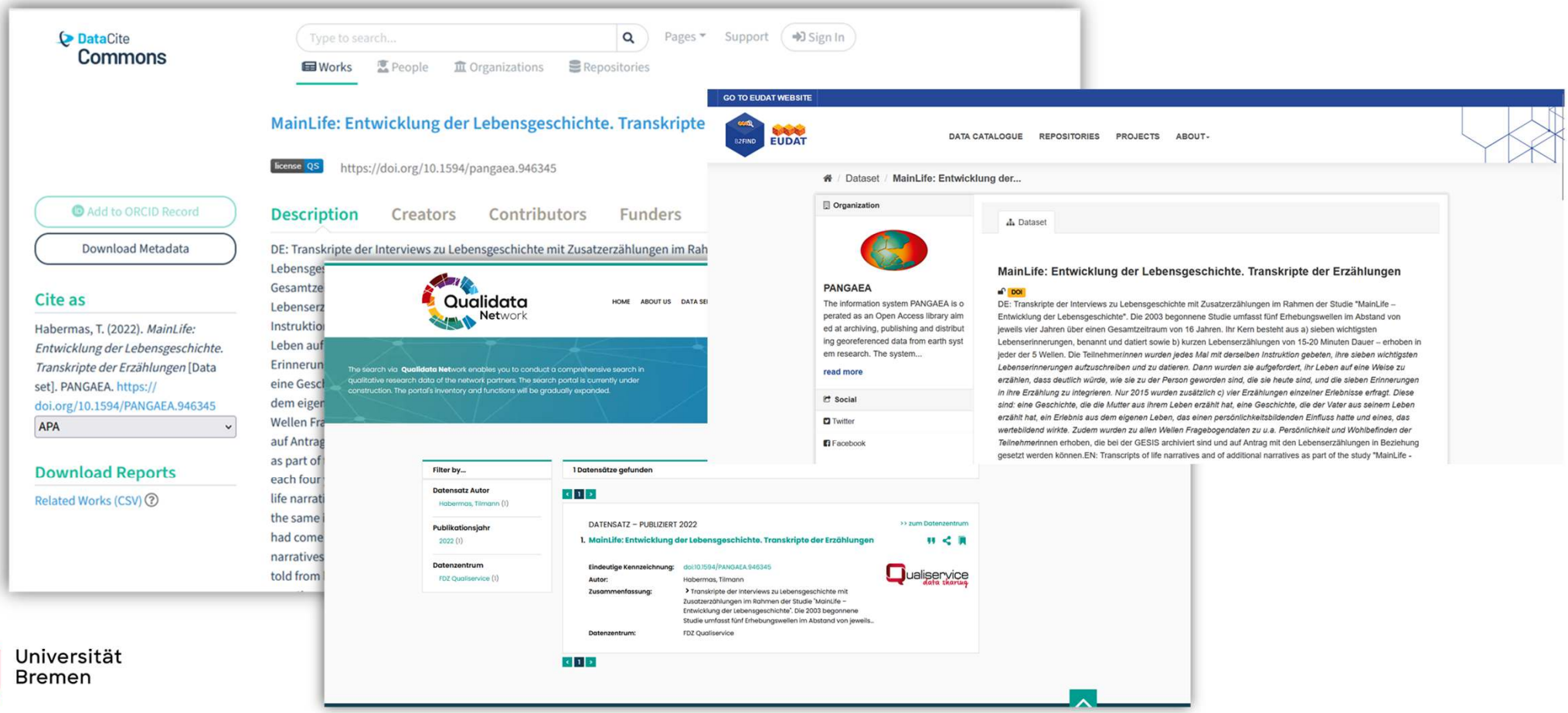
The Qualiservice Metadata-Model

- takes into account the work of the former DDI Alliance Qualitative Data Working Group (Hoyle et al. n.d.)
- takes into account other metadata schemas such as Dublin Core, DCAT and the REFI-QDA standard
- The Qualiservice-metadata form suggests terms for description and leaves room for individual entries
- Project networks and Collaborative Research Centres can be recorded as well as individual qualitative projects and various data types, waves of data collection and methods
- Qualiservice co-operates with GESIS to present mixed-methods studies - reference to the respective project part in the metadata so that the joint study context remains visible
- Contextualisation is supplemented by a published study report, which embeds the data set in its context of origin and is part of the metadata. The study report is searchable in full text.
- “Micrometadata” provide information at case or object level
- enhances the FAIRness of the metadata, which is addressed in the following slides

Findability

- All data records can be read and found internationally on different platforms



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Accessibility

For machines

Rights Management	Access constraints: access rights needed
Rights Management	info:eu-repo/semantics/restrictedAccess

```
<rightsList>
    <rights>Data access is restricted (moratorium, sensitive data, license constraints)</rights>
</rightsList>
```

```
<md:attribute attid="50375" name="Statement">Upon request by Qualiservice</md:attribute>
```

For humans

Show more...

Domain attribute(s):

Access type: OnSite * *Analysis unit:* Individual * *Collection completeness:* DE: Von den 116 Personen aus der Welle 1 zu Messpunkt 1 haben 79 Personen von 116 ihr Einverständnis für die Nutzung ihrer erhobenen Daten für wissenschaftliche Nachnutzung gegeben. Zum Zeitpunkt 2 der Welle 1 sind es 72 Personen von 108. Aus der 2. Welle gaben 119 Personen von 168 ihren Einverständnis ab. In der Welle 3 sind es 119 Personen von 150. Aus der Welle 4 gaben 115 von 135 Personen ihren Einverständnis ab. Aus der Welle 5 sind es 120 Personen von 138. * *Kind of data:* VerbalData.Transcripts * *Statement:* Upon request by Qualiserviece * *Study class:* SecondaryUse.Research * *Sub-universe:* DE: Das Sample von insgesamt 172 Personen umfasst sechs Geburtskohorten. Die 4 jüngeren Kohorten waren zu Beginn ca. 8, 12, 16 und 20 Jahre alt, 2007 wurden zwei ältere Kohorten hinzugefügt, damals im Schnitt 40 und 65 Jahre alt. Beide Geschlechter waren gleichermaßen vertreten. Die jüngste Gruppe wurde unter den besseren SchülerInnen (Lehrerurteil) dritter Klassen einer Grundschule ausgesucht, die Kohorten 2 bis 4 waren SchülerInnen eines Gymnasiums. Die Kohorten 5 und 6 wurden durch Aushänge und die Universität des höheren Lebensalters akquiriert. In Qualiservice vertreten sind nur diejenigen Teilnehmenden, die 2015 oder 2019 teilnahmen und sich nachträglich mit der Verwahrung in Qualiservice einverstanden erklärten. Das sind insgesamt 123 Fälle. * *Title language:* de * *URI:* unihb.eu

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Interoperability

Semantic interoperability

Thesauri


```
<attribute attid="48145" name="Keyword" >Politisches Verhalten (TheSoz: 10040700); Wahlverhalten (TheSoz: 10061173); Russlanddeutsche; German-Russians (Wikidata: Q702193);</attribute>
```


ISO-Codes


```
<attribute attid="47520" name="Language" >deu</attribute>
<attribute attid="50254" name="Location" >Köln; Duisburg; North Rhine-Westphalia (ISO 3166: DE-NW)</attribute>
```


Classifications


```
<attribute attid="44221" name="Subject" >Politikwissenschaft (discipline); Politikwissenschaft (DFG: 111-04); Politics.Elections (CESSDA TC); Politics.PoliticalBehaviourAndAttitudes (CESSDA TC);</attribute>
```


Qualified links to other resources

Supplement to:	**Köber, Christin; Schmiedek, Florian; Habermas, Tilmann (2015):** Characterizing lifespan development of three aspects of coherence in life narratives: A cohort-sequential study. *Developmental Psychology*, **51(2)**, 260-275, https://doi.org/10.1037/a0038668
Related to:	**Habermas, Tilmann (2022):** MainLife – Entwicklung der Lebensgeschichte (quantitativer Teil) [dataset]. *GESIS Data Arcive*, https://doi.org/10.4232/1.13955 *(quantitative data)*
Study report:	**Habermas, Tilmann (2022):** The longitudinal study of brief life narratives: Mainlife Study (2002-2019). Study Report. *Bremen: FDZ Qualiservice, GESIS*, 17 pp, https://doi.org/10.26092/elib/1651

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Reusability

Provenance

Event(s):

MainLife_01 * *Date/Time Start:* 2002-01-01T00:00:00 * *Date/Time End:* 2003-12-31T00:00:00 * *Collection situation:* Es handelt sich um ganze Lebenserzählungen von ca. 15 Minuten Länge, die durch die Teilnehmer*innen selbst strukturiert sind. * *Event type:* DataCollection * *Instrument:* Guidelines.Flex.Narrative * *Methodology:* DE: Lebensgeschichte mit anfänglichem Erzählstimulus * *Mode of collection:* FaceToFace * *Responsible:* Cybèle de Silveira (I-88736); * *Sample size:* 114 * *Sampling procedure:* Die 4 Kohorten waren ca. 8, 12, 16 und 20 Jahre alt. * *Location:* Hesse, Germany * *Campaign:* MainLife (Entwicklung der Lebensgeschichte) * *Method/Device:* Erzählung. Narration (Narration) * *Comment:* DE: Die erste Welle umfasst nur die Kohorten 1 bis 4 (8-20 Jahre). Das Thema war die Entwicklung der Lebensgeschichte. Es gab eine zweifache Erhebung der Lebenserzählung in den Schulen mit 2 Wochen Abstand, um die Stabilität der Erzählung zu testen und den möglichen Einfluss eines „Trainings", das in der ersten der beiden Sitzungen bei der Hälfte der Teilnehmer*innen durchgeführt wurde, die andere bekam eine Kontrollaufgabe. Bei dem Training wurde ein wichtiges Lebensereignis zusätzlich erzählt, zu dem die Erzählenden dann semistrukturiert so befragt wurden, dass sie dazu angeregt wurden, biographische Bezüge temporaler, kausal-motivationaler und thematischer Art herzustellen (unveröffentlicht). Die Kontrollaufgabe bestand aus einer vergleichbar ausführlichen und interaktiven Diskussion über ein Naturphänomen (Wasser). Das Training wirkte sich nicht auf die Kohärenz der Lebenserzählungen aus.

MainLife_02 * *Date/Time Start:* 2007-01-01T00:00:00 * *Date/Time End:* 2007-12-31T00:00:00 * *Collection situation:* DE: Es handelt sich um ganze Lebenserzählungen von ca. 15 Minuten Länge, die durch die Teilnehmer*innen selbst strukturiert sind. * *Event type:* DataCollection * *Instrument:* Guidelines.Flex.Narrative * *Methodology:* DE: Lebensgeschichte mit anfänglichem Erzählstimulus * *Mode of collection:* FaceToFace * *Responsible:* Alexa Grabhorn (I-88737); Verena Diel (I-88739); * *Sample size:* 164 * *Sampling procedure:* Die Kohorten 1-4 waren bei der 2. Welle 12, 16, 20 und 24 Jahre alt * *Location:* Hesse, Germany * *Campaign:* MainLife (Entwicklung der Lebensgeschichte) * *Method/Device:* Erzählung. Narration (Narration) * *Comment:* DE: Bei der zweiten Welle wurden die Kohorten 5-6 hinzugenommen. Das Thema war die Entwicklung der Lebensgeschichte longitudinal.

MainLife_03 * *Date/Time Start:* 2011-01-01T00:00:00 * *Date/Time End:* 2011-12-01T00:00:00 * *Collection situation:* Es handelt sich um ganze Lebenserzählungen von ca. 15 Minuten Länge, die durch die Teilnehmer*innen selbst strukturiert sind. * *Event type:* DataCollection * *Instrument:* Guidelines.Flex.Narrative * *Methodology:* DE: Lebensgeschichte mit anfänglichem Erzählstimulus * *Mode of collection:* FaceToFace * *Responsible:* Christin Camia (I-88756); Anda Constantinescu (I-88760); * *Sample size:* 150 * *Sampling procedure:* Die bisherigen Teilnehmer*innen wurden bis zu drei Mal per Post angeschrieben, dann per Telefon und/ oder E-Mail kontaktiert. * *Location:* Hesse, Germany * *Campaign:* MainLife (Entwicklung der Lebensgeschichte) * *Method/Device:* Erzählung. Narration (Narration) * *Comment:* DE: Das Thema war die Entwicklung der Lebensgeschichte longitudinal, die Stabilität bzw. Veränderung der Lebensgeschichte und Zusammenhänge mit Wohlbefinden und Persönlichkeit.

Change history: 2021-04-06T13:49:47 – Data collection location values from PANGAEA controlled vocabulary

License / User Agreement

QS-Licence / user agreement
Classification of Data: "Secondary use : Research", "Secondary use : Teaching"

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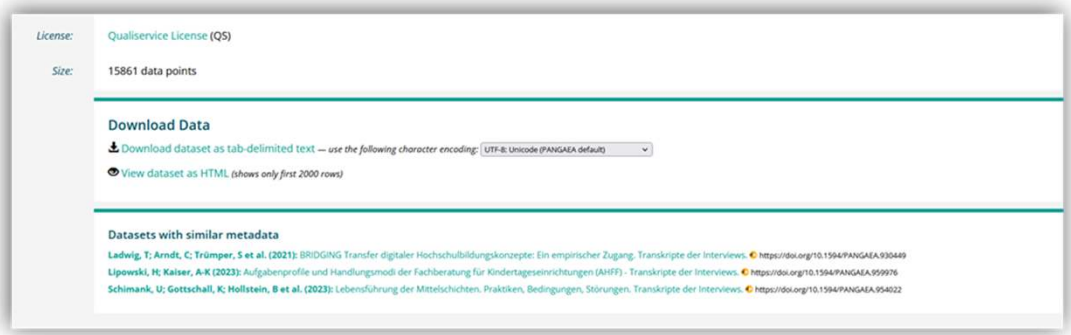
„Micrometadata“ at Qualiservice

- Non-sensitive information about the case itself (e.g. socio-demographic data),
- Information on the data object (duration of the interview, file format, location of the interview etc.),
- Allow an overview of the entire data set at this level of detail,
- Important for assessing the data fit or selecting suitable interviews (as a data set can contain a large number of interviews (over 100 or more)



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Data

Download dataset as tab-delimited text — *use the following character encoding:* UTF-8: Unicode (PANGAEA default)

1 Case ID	2 Data ID	3 File name	4 Data format (Datenformat)	5 Content	6 DC Method	7 Language	8 Event	9 DC date	10 Event duration	11 Gender (Geschlecht)	12 Birth year (Geburtskohorte, min)	13 Birth year (Geburtskohorte, max)	14 Occup. status	15 School grade	16 Birth place
1	MainLife_01-1-LG_1-226	MainLife_IT-LG_01-1_1-226	Text	Lebensgeschichte	Narration	deu	MainLife_01	2003	PT16M5S	female	1993	1995	keine Angabe	keine Angabe	Deutschlanc
1	MainLife_01-2-LG_1-1256	MainLife_IT-LG_01-2_1-1256	Text	Lebensgeschichte	Narration	deu	MainLife_01	2003	PT14M33S	female	1993	1995	keine Angabe	keine Angabe	Deutschlanc
1	MainLife_02-LG_1-980	MainLife_IT-LG_02_1-980	Text	Lebensgeschichte	Narration	deu	MainLife_02	2007	PT11M93S	female	1993	1995	Schule	keine Angabe	Deutschlanc
1	MainLife_03-LG_1-590	MainLife_IT-LG_03_1-590	Text	Lebensgeschichte	Narration	deu	MainLife_03	2011	PT9M07S	female	1993	1995	keine Angabe	Keiner	Deutschlanc
1	MainLife_04-LG_1-141	MainLife_IT-LG_04_1-141	Text	Lebensgeschichte	Narration	deu	MainLife_04	2015	PT12M38S	female	1993	1995	Keine Berufstätigkeit	Keiner	Deutschlanc
1	MainLife_04-ZM_1	MainLife_IT-ZM_04_1	Text	Zusatzerzählung Mutter	Narratives Interview	deu	MainLife_04	2015	PT2M32S	female	1993	1995	Keine Berufstätigkeit	Keiner	Deutschlanc
1	MainLife_04-ZV_1	MainLife_IT-ZV_04_1	Text	Zusatzerzählung Vater	Narratives Interview	deu	MainLife_04	2015	PT2M28S	female	1993	1995	Keine Berufstätigkeit	Keiner	Deutschlanc
1	MainLife_04-ZP_1	MainLife_IT-ZP_04_1	Text	Zusatzerzählung Persönlichkeit	Narratives Interview	deu	MainLife_04	2015	PT1M56S	female	1993	1995	Keine Berufstätigkeit	Keiner	Deutschlanc
1	MainLife_04-ZW_1	MainLife_IT-ZW_04_1	Text	Zusatzerzählung Werte	Narratives Interview	deu	MainLife_04	2015	PT1M55S	female	1993	1995	Keine Berufstätigkeit	Keiner	Deutschlanc
2	MainLife_01-1-LG_2-1046	MainLife_IT-LG_01-1_2-1046	Text	Lebensgeschichte	Narration	deu	MainLife_01	2003	PT9M67S	female	1993	1995	keine Angabe	keine Angabe	Deutschlanc
2	MainLife_01-2-LG_2-1297	MainLife_IT-LG_01-2_2-1297	Text	Lebensgeschichte	Narration	deu	MainLife_01	2003	PT9M45S	female	1993	1995	keine Angabe	keine Angabe	Deutschlanc
2	MainLife_02-LG_2-465	MainLife_IT-LG_02_2-465	Text	Lebensgeschichte	Narration	deu	MainLife_02	2007	PT14M46S	female	1993	1995	Schule	keine Angabe	Deutschlanc
2	MainLife_03-LG_2-131	MainLife_IT-LG_03_2-131	Text	Lebensgeschichte	Narration	deu	MainLife_03	2011	PT10M05S	female	1993	1995	Schule	Mittlere Reife	Deutschlanc
2	MainLife_04-LG_2-1088	MainLife_IT-LG_04_2-1088	Text	Lebensgeschichte	Narration	deu	MainLife_04	2015	PT9M47S	female	1993	1995	Berufstätigkeit	Fachabitur	Deutschlanc
2	MainLife_05-LG_2-3	MainLife_IT-LG_05_2-3	Text	Lebensgeschichte	Narration	deu	MainLife_05	2019	PT18M54S	female	1993	1995	Berufstätigkeit	Abitur	Deutschlanc
2	MainLife_04-ZM_2	MainLife_IT-	Text	Zusatzerzählung	Narratives	deu	MainLife_04	2015	PT2M01S	female	1993	Row 15, column 12	Berufstätigkeit	Fachabitur	Deutschlanc

Habermas, Tilmann (2022): MainLife: Entwicklung der Lebensgeschichte. Transkripte der Erzählungen [dataset]. Qualiservice, PANGAEA, https://doi.org/10.1594/PANGAEA.946345

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2) QualiTerm – a controlled vocabulary for qualitative data

- Is currently being developed by QualidataNet - Network for qualitative data (https://www.qualidatanet.com/en/)
- Objective:
  - To map the heterogeneity of the methods and create a clear language for the description of qualitative data in metadata standards
  - To increase the interoperability and reusability of qualitative data
- QualiTerm facilitates machine readability, and the linking of metadata, because it suggests the use of a common language.

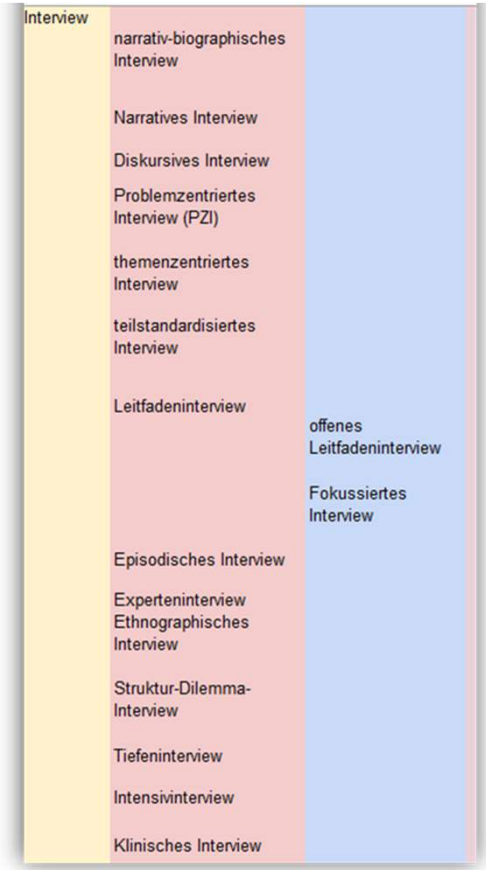
QualiTerm - Controlled Vocabulary for qualitative data

Procedure:

- Identification of gaps in existing vocabularies (thesaurus, DDI vocabularies)
- Collection and organization of terms in a collaborative process involving researchers from different disciplines
- description with scope notes, selection and prioritization of preferred terms
- Focus on : sampling, collection, analysis, processing methods
- Currently: Revision of the scope notes, decisions on open points for practical application, translation into English
- Some RDCs have already partially implemented QualiTerm or used it for orientation

QualiTerm-Element: Collection Methods

- New development in cooperation with researchers from various disciplines
- Currently 26 main categories (including: Interview, observation, photography, simulation, videography, camera ethnography...)

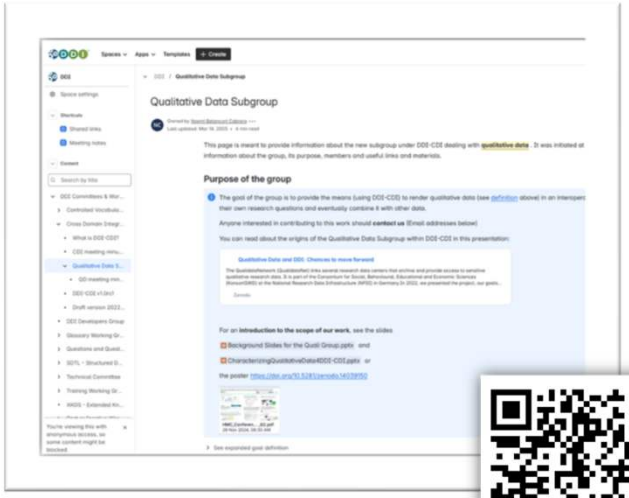


QualiTerm-Element: Sampling Methods

- Identification of existing vocabularies - e.g. DDI-CV Sampling Procedure - identification of gaps, suggestions for supplementing the DDI vocabulary, exchange with DDI-CV Working Group
- Feedback at working level: possibly addition of the term ‘theoretical sampling’ to the DDI vocabulary

3) Participation in international metadata-development

- DDI-CDI Qualitative Data Subgroup, coordinated by QualidataNet
- DDI-CDI was developed specifically for the description of cross-domain data for the purpose of integration.
- Objective: To create an extension of the DDI-CDI model to describe and integrate less structured, non-numeric (qualitative) data types



Working Group Wiki-Site: https://ddi-alliance.atlassian.net/wiki/spaces/DDI4/pages/3083862017/

Use cases of qualitative data to test the DDI-CDI model

- We searched for uise cases from different disciplines
- Purpose of the use cases:

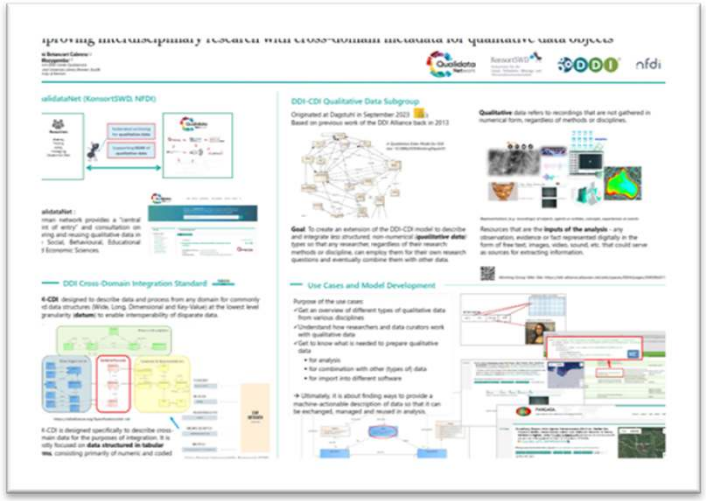
✓ Get an overview of different types of qualitative data from various disciplines

✓ Understand how researchers and data curators work with them

✓ Get to know what is needed to prepare them

  - for analysis
  - for combination with other collected data
  - for import into different software

→ Ultimately, to identify which are the most granular pieces that need to be described



https://zenodo.org/records/14039150

Use Case: Annotaded Video Data

/* DATA DESCRIPTION:
Citation: Gruber, Martin: Traditional Beekeeping and Honey Hunting in Central Cameroon. Audiovisual material. Qualiservice, PANGAEA, https://doi.pangaea.de/10.1594/PANGAEA.963249 (dataset in review)
Abstract: The materials are video recordings made for the ethnographic documentary "Gbaya. Beekeeping and Honey Hunting" (Gruber 2015) in the Adamaoua Region of Cameroon. Ethnographic film was perce
Coverage: DATE/TIME START: 2015-03-01T00:00:00 * DATE/TIME END: 2015-04-30T00:00:00
Event(s): BeeCultures_Footage * DATE/TIME START: 2015-03-01T00:00:00 * DATE/TIME END: 2015-04-30T00:00:00 * Event type: DataCollection * Instrument: Guidelines.Flex * Methodology: The video data
Domain attri Access type: On-site * Kind of data: Observations.Video * Study class: SecondaryUse.Research&Teaching * Title language: eng
Comment: Study-Materials note: The context material includes an interview guide. The data set contains data in the following languages: French, Fulfulde and Gbaya.
License: Qualiservice License (QS)
Size: 1925 data points
*/

Data ID	File name	startOffset (video segments)	endOffset (video segments)	Content (title segments)	Annotation (biologist)	Annotation (anthropologist)	Keywords (keywords segments)
BC_01_01	AA0874_01	00:00:00	00:00:17		early morning traffic noise from the nation highway		
BC_01_63	AA0937_01	00:00:00	00:01:28		He explains where to find rope and the difference between thin and thick rope, as well as their uses		Craft; Beehive material
BC_01_64	AA0941_01	00:00:00	00:29:11		Spraying water to soften the herbs and facilitate braiding	Extreme long shot of the entire process of wrapping the framework with grasses; gives an impresssion of the time necesssary for the hive's contruction	Beehive; Beehive materials; Apprenticeship; Craft; Aesthetics
BC_01_64	AA0941_01	00:00:00	00:05:09	Softening the herbs		Where does the grass come from? Would it possible to use freshgrass?	
BC_01_64	AA0941_01	00:22:00	00:23:51		He talks how beauty and functionality go together in the construction of the hives	Significance of the tip of the hive where functionality and aesthetics unite	Aesthetics
BC_01_64	AA0941_01	00:23:51	00:24:17	Inside the hive			
BC_01_64	AA0941_01	00:24:17	00:25:01	Closing the hive with its lid			
BC_01_64	AA0941_01	00:25:01	00:29:11	Sealing the hive with mud			
BC_01_67	AA0944_01	00:00:00	00:01:07				Beeplants; Landscape
BC_01_68	AA0945_01	00:00:00	00:01:18				Beeplants; Landscape
BC_01_69	AA0946_01	00:00:00	00:00:30				Beeplants; Landscape; Beehives
BC_01_70	AA0947_01	00:00:00	00:00:13		Dry the mud in the sun before placing the hive on the tree		Beehive

Web:
https://www.qualidatanet.com/

Mail:
info@qualidatanet.com

Part of *KonsortSWD, NFDI*
(DFG funding number 442494171)

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